Sharp Phase Budgets for Moment-Calibrated Scalar Wiener–Hopf Factors

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Abstract

A positive scalar kernel can be accurately known in magnitude while its normalized Wiener–Hopf factor remains sensitive in phase. We quantify how much a finite set of calibrated logarithmic moments resolves this ambiguity. For real log-kernel errors supported on a band, bounded by ε, and annihilating polynomials through degree $2r$, the exact worst-case unwrapped phase error at normalized height y above the band centre is $\varepsilon\pi^{-1}\log\coth((r+1)\operatorname{arsinh} y)$. The corresponding sharp log-magnitude error is $2\varepsilon\pi^{-1}\arctan\exp(-(2r+3)\operatorname{arsinh} y)$. Explicit Chebyshev sign functions attain both bounds, separately. The phase law yields an exact calibration-order rule, a boundary-resolution transition, and a closed sensitivity formula when one moment is inexact. Weighted and off-centre uncertainty reduces to classical best L^1 approximation. A certificate-producing implementation combines finite linear programming, exact moment repair, rational root isolation, and ball arithmetic to bracket the continuum problem. Computations include 312 formula checks, 4,056 moment checks, 28 continuum brackets, and smooth near-extremizers. The contribution is a sharp uncertainty calculation for a declared scalar factorization model, with reproducible software for calibration and numerical-method assessment.

Keywords: Wiener–Hopf factorization; spectral factorization; Cauchy transform; logarithmic moments; best L^1 approximation; validated numerics.

MSC 2020: 30E25, 41A10, 65G20.

Software and computational records: https://github.com/shi1720/phasebudgets.

1 Introduction

Scalar Wiener–Hopf factorization separates a kernel into factors analytic in opposite half-planes. In numerical applications the kernel is rarely known exactly: it may be approximated, reconstructed, or calibrated from a model. An error bound for the supplied kernel is then only the beginning of an accuracy assessment. Which uncertainty affects factor magnitude, which affects phase, and how much additional calibration is needed at a specified complex point?

The issue is visible even for positive kernels, where index changes and logarithmic branch selection on the real axis are absent. An upper-half-plane factor is the exponential of a Cauchy transform of the real logarithm. Its log-magnitude is a Poisson average, while its unwrapped phase is a conjugate Poisson average. The former has a bounded averaging kernel; the latter becomes singular near the real axis. The resulting distinction is classical. Our question concerns the exact improvement available from a prescribed finite set of integral constraints on the log-kernel error.

Suppose uncertainty is confined to $[-1,1]$. At $z=iy$, assume

$$|h(t)|\le\varepsilon,\qquad \int_{-1}^{1} t^j h(t)\,dt=0,\quad j=0,\dots,2r.$$

These are *logarithmic* calibration moments: h is the logarithm of the ratio of two positive kernels. They are neither covariance moments of the kernel nor samples of the factor. Under exactly these assumptions, we obtain separate sharp phase and log-magnitude budgets. The extremizers have explicitly known switches. Their discontinuities are not essential to the supremum: smooth, compactly supported perturbations approach the same budgets while retaining every calibration moment.

Three findings organize the study. The budgets have elementary closed forms involving $(r+1)\operatorname{arsinh} y$ and $(2r+3)\operatorname{arsinh} y$. Increasing moment order gives exponential improvement at fixed height, but any fixed order retains logarithmic phase sensitivity as $y \downarrow 0$. Finally, a finite-dimensional search can supply candidate optimizers while separate continuum arguments certify their accuracy.

The intended use is an accuracy interface around an existing scalar factorization routine. A practitioner supplies a justified log-error envelope, calibration constraints, and evaluation location. The software returns a budget or a replayable bracket. We do not infer those assumptions from a small residual, claim a new general factorization algorithm, or assert that a factor error alone bounds a complete physical prediction.

1.1 Relation to earlier work

The Wiener–Hopf method and constructive approximations have a broad literature; Kisil, Abrahams, Mishuris, and Rogosin survey its analytical and numerical development [1]. Kisil gives scalar approximation and factor-error estimates in L^p, $1 < p < \infty$ [2]. Our uncertainty class combines a pointwise log-error envelope with exact finite-band moments and asks for a sharp pointwise error of the normalized analytic logarithm.

Gain–phase sensitivity is established. Anderson and Green obtain degree-dependent bounds for rational functions [3]. Sandberg, Lanzon, and Anderson use magnitude and phase criteria in approximation and identification [4]. Boche and Pohl analyze supremum-norm discontinuity and finite-dimensional error growth in spectral factorization [5]. Ephremidze, Shargorodsky, and Spitkovsky give quantitative scalar H^2 continuity estimates involving both spectrum and logarithm [6]. The present boundary-sensitivity result is not a new discovery of factorization instability.

Additional calibration information has a related role in subtractive dispersion relations. Palmer, Williams, and Budde use independent reference values to improve finite-band Kramers–Kronig reconstruction [7]. Our constraints are integrals of a bounded log-error on its uncertain band, rather than point anchors or a model for omitted tails.

The mechanism is classical best L^1 approximation and annihilating sign functions. Hobby and Rice study moment annihilation in L^1 approximation [8]; Nakatsukasa and Townsend discuss the special role of second-kind Chebyshev nodes and numerical best L^1 approximation [9]. We use these established tools explicitly. The contribution is their specialization to an exact factor-uncertainty law, its calibration and inexact-moment consequences, and a continuum certification workflow. This specialization is distinct from the general approximation identities and instability results on which it rests.

2 A normalized scalar uncertainty model

Let $K_0(t) > 0$ be a reference kernel with a chosen upper-half-plane factor $K_{0,+}$. Consider

$$K(t) = K_0(t)e^{h(t)}, \qquad h : \mathbb{R} \to \mathbb{R}, \qquad \operatorname{supp} h \subseteq [-1, 1]. \tag{1}$$

For bounded measurable h, define

$$\mathcal{C}h(z) = \frac{1}{2\pi i}\int_{-1}^{1} \frac{h(t)}{t - z}\, dt, \qquad \operatorname{Im} z > 0. \tag{2}$$

The upper factor ratio and its normalized analytic logarithm are

$$R_+(z) := \frac{K_+(z)}{K_{0,+}(z)} = \exp(\mathcal{C}h(z)), \qquad \log R_+(z) := \mathcal{C}h(z). \tag{3}$$

The normalization is $\mathcal{C}h(z) \to 0$ at infinity. Its imaginary part is an *unwrapped* phase, not restricted to $(-\pi, \pi]$. The Plemelj boundary relation gives $2\operatorname{Re}\mathcal{C}h(t+i0) = h(t)$ almost everywhere, so $|R_+(t+i0)|^2 = e^{h(t)}$. This is positive scalar spectral factorization on the line even if K has no analytic continuation through the band. We do not assume strip analyticity for the discontinuous extremal kernels.

At $z = x + iy$,

$$\begin{aligned} \operatorname{Re}\mathcal{C}h(z) &= \frac{1}{2\pi}\int_{-1}^{1} \frac{yh(t)}{(t-x)^2+y^2}\,dt, \\ \operatorname{Im}\mathcal{C}h(z) &= -\frac{1}{2\pi}\int_{-1}^{1} \frac{(t-x)h(t)}{(t-x)^2+y^2}\,dt. \end{aligned} \tag{4}$$

The main uncertainty class is

$$\mathcal{H}_m(\varepsilon) = \left\{ h \in L^\infty([-1,1];\mathbb{R}) : \|h\|_\infty \le \varepsilon, \quad \int_{-1}^{1} p(t)h(t)\,dt = 0 \quad \forall p \in \mathbb{P}_m \right\}. \tag{5}$$

Any basis of $\mathbb{P}_m$ expresses the same constraints. Numerically we use Legendre moments.

Remark 2.1 (Calibration and symmetry). A measured spectrum need not provide exact logarithmic moments. The class (5) is a supplied deterministic model; inexact moments are treated in Section 4. Moreover, a positive scalar spectrum need not be even. If both kernels must be even, their relative logarithm is even and the phase error at iy is identically zero. Off-centre evaluation or a different symmetry-constrained class is then required.

For a band $[\omega_c - W, \omega_c + W]$, set $t = (\omega - \omega_c)/W$ and $z = (\zeta - \omega_c)/W$. The Cauchy transform is invariant under this affine change. Thus y is the physical imaginary height divided by the half-bandwidth, and the calibration moments refer to the normalized coordinate.

3 Exact budgets at the band centre

Define

$$B_r(y) = \sup_{h \in \mathcal{H}_{2r}(\varepsilon)} |\operatorname{Im}\mathcal{C}h(iy)|, \qquad A_r(y) = \sup_{h \in \mathcal{H}_{2r}(\varepsilon)} |\operatorname{Re}\mathcal{C}h(iy)|.$$

Theorem 3.1 (Sharp phase and log-magnitude laws)**.** *For an integer $r \ge 0$, $y > 0$, and $\varepsilon \ge 0$,*

$$B_r(y) = \frac{\varepsilon}{\pi} \log\coth\left((r+1)\operatorname{arsinh} y\right), \tag{6}$$

$$A_r(y) = \frac{2\varepsilon}{\pi} \arctan\left(e^{-(2r+3)\operatorname{arsinh} y}\right). \tag{7}$$

For $\varepsilon > 0$, the integral $\int th(t)/(t^2+y^2)\,dt$ is uniquely maximized, up to null sets, by

$$h_B(t) = \varepsilon\, \operatorname{sgn}\big((-1)^r U_{2r+1}(t)\big). \tag{8}$$

The positive real part is uniquely maximized by

$$h_A(t) = \varepsilon\, \operatorname{sgn}\big((-1)^{r+1} U_{2r+2}(t)\big). \tag{9}$$

Here U_n is the Chebyshev polynomial of the second kind. The witnesses do not depend on y.

3.1 Two approximation identities

The annihilating-sign structure in the following identities belongs to classical L^1 approximation [8, 9]. We give the needed proofs.

Lemma 3.2 (Canonical signs)**.** *For $n \geq 1$, $\sigma_n(t) = \operatorname{sgn} U_n(t)$ annihilates $\mathbb{P}_{n-1}$ in unweighted integration on $[-1, 1]$.*

Proof. Set $t = \cos\theta$, $0 < \theta < \pi$. Since $U_n(\cos\theta) = \sin((n+1)\theta)/\sin\theta$,

$$\int_{-1}^{1} p(t)\sigma_n(t)\,dt = \int_0^{\pi} p(\cos\theta)\sin\theta\,\operatorname{sgn}\sin((n+1)\theta)\,d\theta.$$

For $\deg p < n$, $p(\cos\theta)\sin\theta$ expands into sine modes of frequencies at most n. The square wave has only odd multiples of $n+1$ in its L^2 sine series. Orthogonality proves the assertion. □

Let $w(z) = z - \sqrt{z^2-1}$ off $[-1, 1]$, choosing the branch with $w(z) \to 0$ at infinity. In the upper half-plane $|w| < 1$; at iy, $w = -ie^{-\operatorname{arsinh} y}$.

Lemma 3.3 (Transform of a canonical sign)**.** *With the logarithm continuous from infinity,*

$$\int_{-1}^{1} \frac{\sigma_n(t)}{t-z}\,dt = -2\log\frac{1+w(z)^{n+1}}{1-w(z)^{n+1}}. \tag{10}$$

The ratio on the right has positive real part for $|w| < 1$, so the principal logarithm gives the prescribed branch.

Proof. The left side is analytic off the segment, tends to zero at infinity, and has upper-minus-lower jump $2\pi i\sigma_n$ away from switch points. For $t = \cos\theta$ approached from above, $w = e^{-i\theta}$ and

$$\frac{1+e^{-i(n+1)\theta}}{1-e^{-i(n+1)\theta}} = -i\cot\frac{(n+1)\theta}{2}.$$

The imaginary part of the right side's upper trace is therefore $\pi\operatorname{sgn}\sin((n+1)\theta) = \pi\sigma_n(t)$. The lower trace is its conjugate. The two sides have the same jump and decay, with at most logarithmic singularities at endpoints and switches. Their difference extends across the open pieces of the segment. The remaining isolated logarithmic-growth singularities are removable for this single-valued difference. Liouville's theorem proves equality. □

3.2 Proof of the sharp laws

Proof of Theorem 3.1. The case $\varepsilon = 0$ is immediate. For phase, set $n = 2r+1$ and

$$c = \frac{iy}{U_n(iy)}, \qquad p(t) = \frac{t - cU_n(t)}{t^2+y^2}.$$

The number c is real with sign $(-1)^r$. The numerator vanishes at $\pm iy$, so p is real of degree at most $2r-1$; for $r = 0$ it is zero. For every feasible h,

$$\int_{-1}^{1} \frac{th(t)}{t^2+y^2}\,dt = \int_{-1}^{1}\left(\frac{t}{t^2+y^2} - p(t)\right)h(t)\,dt \leq \varepsilon\int_{-1}^{1}\frac{|cU_n(t)|}{t^2+y^2}\,dt. \tag{11}$$

Lemma 3.2 makes h_B feasible, and it attains the inequality pointwise. The residual is nonzero almost everywhere, proving uniqueness.

For gain, set $n = 2r+2$, $c = y/U_n(iy)$, and $p(t) = (y - cU_n(t))/(t^2+y^2)$. This is real of degree at most $2r$, with $\operatorname{sgn} c = (-1)^{r+1}$. The same argument proves optimality and uniqueness of h_A.

Put $a = \operatorname{arsinh} y$ and $\rho = e^{-a}$. For phase, $w^{2r+2} = (-1)^{r+1}\rho^{2r+2}$. Equation (10), multiplied by $(-1)^r$, gives $\operatorname{Im}\mathcal{C}h_B(iy) = -\varepsilon\pi^{-1}\log\coth((r+1)a)$. For gain, $w^{2r+3} = (-1)^r i\rho^{2r+3}$. Using $\log((1+iq)/(1-iq)) = 2i\arctan q$ for real $|q| < 1$ gives (7). Negating the witnesses attains the opposite signs. □

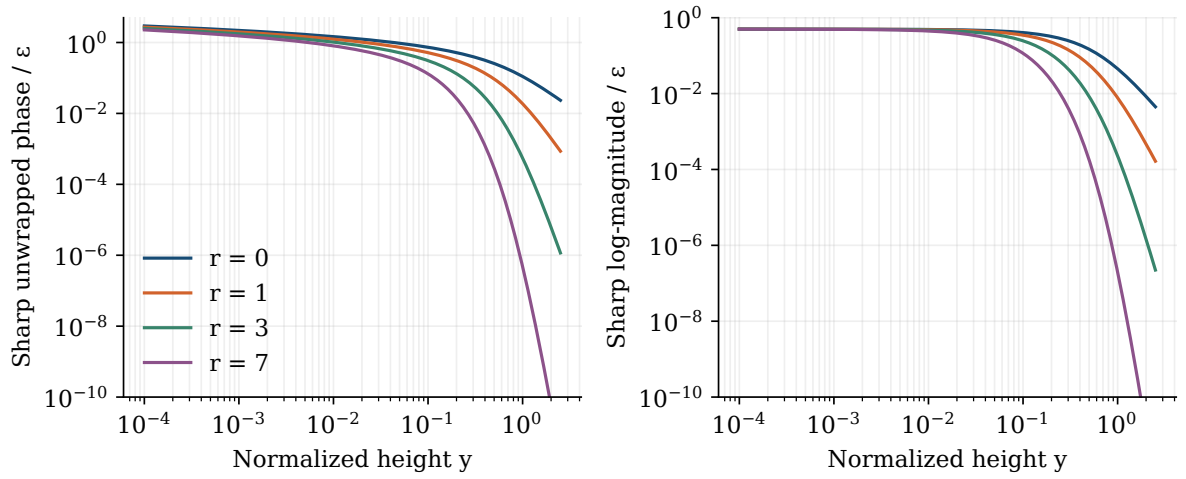


Figure 1: Exact marginal budgets for a unit log-error envelope. Increasing calibration order reduces both at fixed height. At fixed order, phase diverges logarithmically toward the band centre, whereas log-magnitude tends to 1/2.

3.3 Resolution and calibration order

Corollary 3.4 (Boundary resolution)**.** *For fixed r and $y \downarrow 0$,*

$$B_r(y) = \frac{\varepsilon}{\pi} \log \frac{1}{(r+1)y} + o(1), \qquad A_r(y) \longrightarrow \frac{\varepsilon}{2}.$$

For fixed $y > 0$ and $r \to \infty$, writing $a = \operatorname{arsinh} y$,

$$B_r(y) \sim \frac{2\varepsilon}{\pi} e^{-2(r+1)a}, \qquad A_r(y) \sim \frac{2\varepsilon}{\pi} e^{-(2r+3)a}.$$

If $r \to \infty$, $y \downarrow 0$, and $(r+1)y \to \lambda \in (0, \infty)$, then $B_r(y) \to \varepsilon\pi^{-1} \log\coth \lambda$.

Proof. Use $\operatorname{arsinh} y \sim y$, $\coth s \sim 1/s$ at zero, $\log\coth s \sim 2e^{-2s}$ at infinity, and $\arctan s \sim s$ at zero. □

For $\varepsilon, \tau > 0$, the least nonnegative integer order with $B_r(y) \le \tau$ is

$$r_{\min} = \max\left\{0, \left\lceil \frac{\operatorname{artanh}(e^{-\pi\tau/\varepsilon})}{\operatorname{arsinh} y} \right\rceil - 1\right\}. \tag{12}$$

This is a rule for the specified polynomial-moment family, not a lower bound for all calibration measurements. A measurement equal to the phase functional could eliminate its uncertainty directly. For example, with $\varepsilon = 0.02$ and target $\tau = 0.001$, the rule requires $r = 127$ (255 moments) at $y = 0.01$, but only $r = 12$ (25 moments) at $y = 0.1$. Near an integer threshold, evaluate (12) with interval arithmetic rather than an unverified floating-point ceiling.

Parity explains which added moments help. Phase uses only odd h, so the degree-$2r$ condition does not improve its budget beyond degree $2r - 1$ for $r \ge 1$. Gain uses only even h; degrees through $2r + 1$ give the same gain budget as degrees through $2r$.

3.4 The joint set is smaller than its marginal rectangle

The set $\mathcal{Z}_m(z) = \{\mathcal{C}h(z) : h \in \mathcal{H}_m(\varepsilon)\}$ is compact, centrally symmetric, and convex. Weak-* compactness of the feasible set and integrability of the evaluation kernels give compactness. At iy, it lies in $[-A_r, A_r] + i[-B_r, B_r]$ when $m = 2r$. The unique gain maximizer is even, with zero

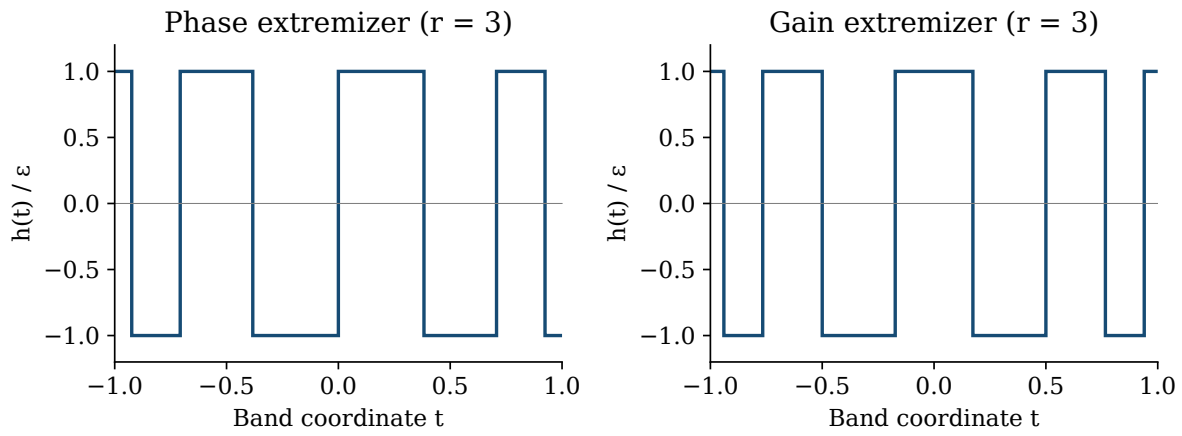


Figure 2: Attaining perturbations with moments through degree six fixed. The phase witness is odd and the gain witness even. Their switches are second-kind Chebyshev roots. The classical annihilating-sign identity supplies feasibility; Theorem 3.1 establishes their extremal factor error.

centre phase; the unique phase maximizer is odd, with zero centre gain. Since both budgets are positive for $\varepsilon > 0$, no corner of that rectangle is attainable. Two separately sharp bounds therefore do not give a jointly sharp complex-error rectangle. Figure 3 illustrates the geometry; directional continuum bounds are developed in Section 5.

4 Inexact calibration and smooth perturbations

Theorem 4.1 (An inexact first moment)**.** *Let $\|h\|_\infty \le \varepsilon$ on $[-1,1]$ with $\varepsilon > 0$, $\int h = \int t^2 h = 0$, and $|\int th| \le \delta$. Set $d = \min(\delta/\varepsilon, 1)$ and $a^2 = (1+d)/2$. The exact phase budget at iy is*

$$B(y,\delta) = \frac{\varepsilon}{2\pi} \log \frac{(y^2+a^2)^2}{y^2(1+y^2)}. \tag{13}$$

For $0 \le \delta < \varepsilon$ an attaining positive phase-integral witness is $h(t) = \varepsilon \operatorname{sgn}(t) \operatorname{sgn}(a^2 - t^2)$. For $\delta \ge \varepsilon$ the constraint is inactive and $\varepsilon \operatorname{sgn} t$ attains it.

Proof. The witness is odd, and its first moment is $\varepsilon(2a^2 - 1) = \min(\delta, \varepsilon)$. For $\delta < \varepsilon$ choose $\lambda = (a^2 + y^2)^{-1}$. The residual

$$\frac{t}{t^2+y^2} - \lambda t = \frac{t(a^2 - t^2)}{(t^2+y^2)(a^2+y^2)}$$

has the witness's sign. Thus

$$\int \frac{th}{t^2+y^2} \le \varepsilon \int \left| \frac{t}{t^2+y^2} - \lambda t \right| + \lambda\delta,$$

with equality for that witness. For $\delta \ge \varepsilon$ use $\lambda = 0$. Integrating on $[0,a]$ and $[a,1]$ gives (13). □

At $\delta = 0$ this is $B_1(y)$, while at $\delta = \varepsilon$ it is $B_0(y)$. The one-sided derivative at exact calibration is

$$\left.\frac{\partial B}{\partial \delta}\right|_{\delta=0} = \frac{1}{2\pi(y^2 + 1/2)}.$$

The derivative stays finite as $y \downarrow 0$: the first moment changes a finite part of the budget but does not remove its boundary logarithm.

Proposition 4.2 (Smooth perturbations have the same suprema)**.** *For fixed finite m and $\operatorname{Im} z > 0$, restricting $\mathcal{H}_m(\varepsilon)$ to real C^∞ functions compactly supported in $(-1,1)$ does not change the supremum of any real linear combination of $\operatorname{Re}\mathcal{C}h(z)$ and $\operatorname{Im}\mathcal{C}h(z)$.*

Proof. Approximate a feasible h by $h_\nu \in C_c^\infty(-1,1)$ with $\|h_\nu\|_\infty \le \varepsilon$ and $h_\nu \to h$ in L^1, using cutoff and mollification. Its moment errors tend to zero. Choose a nonnegative smooth bump η supported in the interval and positive on an open subinterval. For a basis $p_0, \dots, p_m$, the matrix $G_{jk} = \int \eta p_j p_k$ is positive definite. Subtract $\eta \sum c_k p_k$ with $Gc = (\int p_j h_\nu)_j$. This repairs the moments exactly; $c \to 0$ and the correction's supremum norm tends to zero. For $\varepsilon > 0$, divide by $1 + \|\eta \sum c_k p_k\|_\infty / \varepsilon$. The resulting functions are feasible, smooth, and converge to h in L^1. For fixed $\operatorname{Im} z > 0$, both evaluation kernels are bounded. The case $\varepsilon = 0$ is immediate. □

This is smoothness without a fixed derivative bound. A prescribed derivative or bandwidth restriction changes the uncertainty class and may improve its budget.

5 Weighted and off-centre budgets

For a direction $(u, v) \in \mathbb{R}^2$ define

$$k_{u,v,z}(t) = \frac{uy - v(t-x)}{(t-x)^2 + y^2}.$$

Then $u \operatorname{Re}\mathcal{C}h + v \operatorname{Im}\mathcal{C}h = (2\pi)^{-1} \int k_{u,v,z} h$. Let $b(t) > 0$ almost everywhere be integrable, let $p_0, \dots, p_m$ be a basis of $\mathbb{P}_m$, and let $\delta_j \ge 0$. Consider real h with $|h| \le \varepsilon b$ and $|\int p_j h| \le \delta_j$.

Theorem 5.1 (Directional best-L^1 formula)**.** *For $\varepsilon > 0$, $y > 0$, and a nonzero direction, the exact support value is*

$$\sup_h \frac{1}{2\pi} \int k_{u,v,z} h = \frac{1}{2\pi} \min_{\lambda \in \mathbb{R}^{m+1}} \left\{ \varepsilon \int_{-1}^{1} b(t) \left| k_{u,v,z}(t) - \sum_{j=0}^{m} \lambda_j p_j(t) \right| dt + \sum_{j=0}^{m} \delta_j |\lambda_j| \right\}. \tag{14}$$

For exact moments the penalty vanishes; with $b = 1$ this is ordinary best L^1 polynomial approximation.

Proof. Subtract $p = \sum \lambda_j p_j$ inside the integral. The envelope and moment bounds prove the right side is an upper bound for every λ. On the polynomial space, $\int b|p|$ is a norm. The objective is coercive and has a minimizer λ_*. For a nonzero direction, $k - p_*$ is a nonzero rational function with only finitely many zeros on the band. Differentiation of the integral term is justified by domination. Putting $s(t) = \operatorname{sgn}(k(t) - p_*(t))$, subgradient optimality gives

$$\varepsilon \int b(t) s(t) p_j(t)\, dt \in \delta_j \partial |\lambda_{*,j}|.$$

Hence $h_* = \varepsilon b s$ satisfies the moment bounds and attains equality in both the moment and pointwise absolute-value bounds. □

This is a specialization of established L^1 duality. Its value here is computational: any candidate polynomial gives a continuum upper bound when its absolute residual is correctly integrated. With exact moments and $b = 1$, the residual numerator has degree at most $m + 2$, so an optimal witness has at most $m + 2$ interior switches. Off centre, one canonical Chebyshev sign generally does not solve every direction. Certified directional half-planes give an outer polygon for $\mathcal{Z}_m(z)$, while feasible witnesses give an inner set.

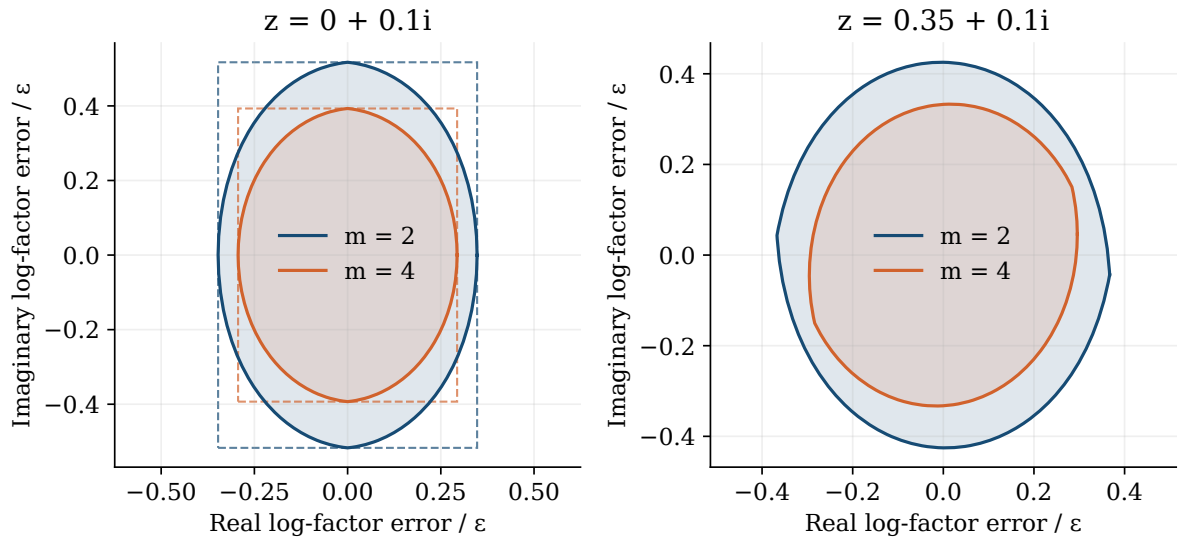


Figure 3: Joint uncertainty for two moment orders. Solid curves are numerical LP candidate polygons with 512 cells and 97 sampled directions, illustrating geometry rather than asserting exact continuum boundaries. Dashed centre rectangles use the exact marginal budgets and have unattainable corners. Separate continuum brackets appear in Table 2.

6 Continuum certificates from finite searches

The reference implementation, `PhaseBudgets`, separates optimization from validation. The general certifier handles a constant envelope, exact Legendre moments, rational x, y, and rational directions. It does not implement every weighted or noisy case of Theorem 5.1.

6.1 Search and exact moment repair

Partition the band into N rational cells and take h_0 constant on each. A linear program maximizes the functional with $-1 \le h_0 \le 1$ and exactly integrated Legendre moment equations. Kernel integrals on cells use logarithms and arctangents. HiGHS, through SciPy, supplies cell values and dual multipliers. These floating-point values are only proposals.

The certifier interprets every candidate decimal as an exact rational number, clips heights to the declared box, and recomputes moments exactly. For Legendre polynomials $P_j(1) = 1$, set

$$\eta_j = \int_{-1}^{1} P_j h_0, \qquad q(t) = \sum_{j=0}^{m} \frac{2j+1}{2} \eta_j P_j(t), \qquad D = \sum_{j=0}^{m} \frac{2j+1}{2} |\eta_j|.$$

Legendre orthogonality and $|P_j| \le 1$ imply that

$$\widehat{h}(t) = \frac{h_0(t) - q(t)}{1 + D} \tag{15}$$

has zero calibration moments and supremum norm at most one. Its directional integral is a lower certificate. Solver feasibility tolerances are therefore not mistaken for exact feasibility.

6.2 Root-isolated integration

The proposed dual is converted to a rational polynomial p. The numerator of $k - p$ is

$$N(t) = uy - v(t - x) - ((t - x)^2 + y^2)p(t).$$

r	Moments	$B_r(0.01)/\epsilon$	$B_r(0.1)/\epsilon$	$A_r(0.1)/\epsilon$
0	1	1.465887	0.734519	0.406060
1	3	1.245283	0.517020	0.347324
3	7	1.024775	0.308511	0.246167
7	15	0.804648	0.130678	0.115349
12	25	0.651214	0.047577	0.042911

Table 1: Exact-law values rounded to six decimals. The moment count includes every degree from zero through $2r$. Phase and gain columns have different attaining perturbations.

SymPy isolates all its real band roots in rational intervals of width at most 10^{-25}. Between these boxes the sign is determined by an exact rational sample, and the integral of $|k-p|$ is a signed antiderivative difference. A root interval of width ℓ contributes at most

$$\ell \frac{\sum_j |N_j|}{y^2}, \qquad N(t) = \sum_j N_j t^j.$$

This follows from $|t| \le 1$ and the denominator bound y^2.

All logarithms, arctangents, and endpoint evaluations use Arb ball arithmetic [10]. Integrals of polynomial multiples of k use exact polynomial division followed by polynomial, logarithmic, and arctangent terms. Records retain rational root boxes and ball enclosures. Exported floating-point endpoints are enlarged by one unit in the last place; the recorded balls retain higher precision.

An inaccurate optimizer consequently leaves a wider bracket rather than invalidating it. A finite cell grid is never used by itself as a continuum upper bound. The interval result still depends on the correctness of its algebraic and ball-arithmetic implementation.

7 Computational study

The deterministic synthetic experiments test the formulas, continuum certification, and smooth near-attainment. They are not measurements of deployed systems. Source, records, figure generation, dependency versions, and automated checks accompany the manuscript.

7.1 Closed forms and witnesses

We use $r = 0, \dots, 12$ and $y \in \{0.001, 0.01, 0.1, 0.3, 1, 3\}$. Both formulas are compared with piecewise antiderivative evaluations at 192 and 384 bits, giving 312 comparisons. The witnesses also give 4,056 ball checks of zero moments through degree $2r$. Every difference or moment enclosure contains zero. The proof of the identities is Theorem 3.1; these computations corroborate their implementation.

A separate mpmath implementation directly integrates each switch interval for 12 phase cases: $r \in \{0, 1, 3, 6\}$, $y \in \{0.01, 0.2, 1\}$. At 70 decimal digits, each absolute difference is below 10^{-60}. These are high-precision numerical comparisons rather than interval proofs. An additional symbolic script verifies 14 polynomial-division identities exactly and checks the canonical-sign transform at 15 off-axis points by separate 65-digit quadrature; the largest transform discrepancy is below 7.47×10^{-64}.

At $y = 0.01$, increasing the number of moments from one to 25 reduces the normalized phase budget from 1.465887 to 0.651214. At $y = 0.1$ it decreases from 0.734519 to 0.047577. These values illustrate the height-dependent transition, rather than uniform removal of boundary sensitivity.

m	x	y	Direction	Lower	Upper
2	0	0.1	$(0,1)$	0.517001	0.517020
2	0.35	0.1	$(0,1)$	0.425340	0.425472
2	0.35	0.1	$(1,1)$	0.547879	0.547967
2	0.35	0.5	$(0,1)$	0.052280	0.052294
4	0	0.1	$(0,1)$	0.393075	0.393125
4	0.35	0.1	$(0,1)$	0.332960	0.333064
4	0.35	0.1	$(1,1)$	0.462509	0.462586
4	0.35	0.5	$(0,1)$	0.019082	0.019094

Table 2: Continuum support brackets for a unit envelope. Direction (u, v) means $u\,\mathrm{Re}\,\mathcal{C}h + v\,\mathrm{Im}\,\mathcal{C}h$; $(1,1)$ is not a unit direction. Centre phase values agree with the closed laws.

7.2 Continuum brackets

We test $m \in \{2,4\}$, $(x,y) \in \{(0,0.1),(0.35,0.1),(0.35,0.5)\}$, and directions $(1,0),(0,1),(1,1)$: 18 certificates with 256 cells. Each repaired primal lower bound lies below its continuum dual upper bound. The largest normalized gap is below 1.33×10^{-4}. Table 2 gives selected brackets with outward rounding.

A refinement study uses 32, 64, 128, 256, and 512 cells for phase at $0 + 0.1i$ and $0.35 + 0.1i$, with $m = 4$, adding ten certificates. The recorded gaps decrease along these sequences. Automated checks also replay four off-centre directions at 192 and 384 bits. The general algorithm is a certificate-producing reference method; no computational optimality or speedup over specialist L^1 solvers is claimed.

7.3 Smoothness and calibration noise

For a numerical realization of Proposition 4.2, use

$$\eta_\kappa(t) = \begin{cases} \exp[-1/(\kappa(1-t^2))], & |t| < 1, \\ 0, & |t| \geq 1, \end{cases} \qquad h_0(t) = \eta_\kappa(t)\tanh\big(\kappa(-1)^r U_{2r+1}(t)\big).$$

Subtract $\eta_\kappa \sum c_j P_j$ using the bump-weighted Gram system, then divide by $1 + \sum |c_j|$. This defines exactly feasible functions when its integrals are exact. The experiment uses adaptive quadrature and a dense numerical solve; it does not validate that Gram solve.

For $r \in \{1,3\}$, $\kappa \in \{8,32,128,512\}$, and $y \in \{0.01, 0.1\}$ there are 16 cases. The largest numerically recomputed moment residual is below 3.76×10^{-16}. Figure 4 shows increasing attainment of the sharp budget. These results do not establish a law under a fixed derivative bound.

An independently assembled 4,096-cell LP checks Theorem 4.1 at $y = 0.2$ for $\delta/\varepsilon \in \{0, 0.1, 0.5, 1\}$. Differences from the normalized formula are below 10^{-6}. The theorem establishes sharpness; the LP provides a numerical cross-check.

7.4 A factor and inverse-factor example

Take the positive reference symbol

$$K_0(t) = \frac{t^2+4}{t^2+1}, \qquad K_{0,+}(z) = \frac{z+2i}{z+i}.$$

The upper factor is analytic and zero-free in the upper half-plane and tends to one at infinity. For $K = K_0 e^{h_B}$, the phase witness gives

$$\frac{K_+(iy)}{K_{0,+}(iy)} = e^{-iB_r(y)}, \qquad e^{-\varepsilon} \leq K/K_0 \leq e^{\varepsilon}.$$

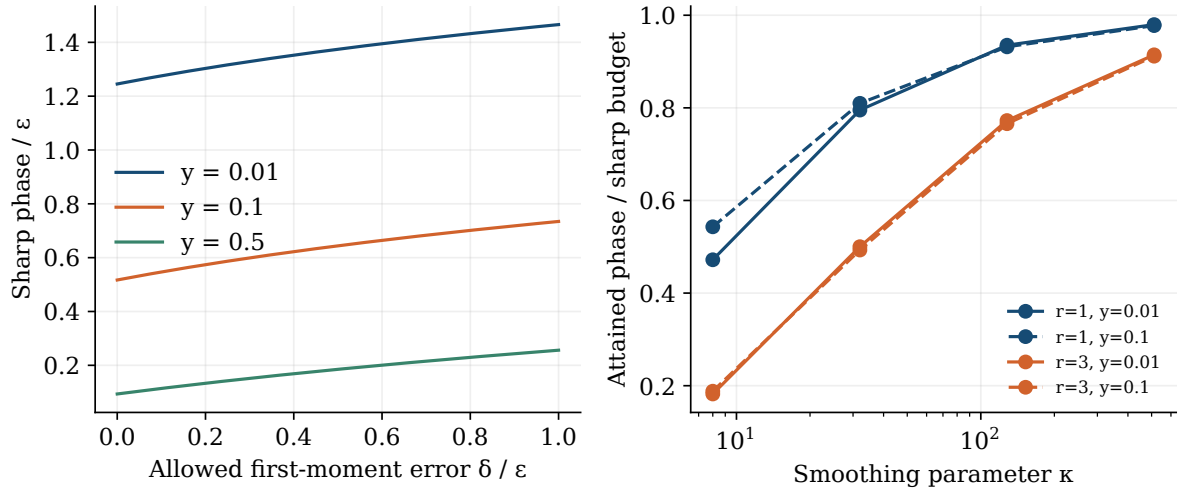


Figure 4: Left: the exact inexact-first-moment law, with mean and second moment fixed. Right: numerical phase attainment after smooth moment repair and norm rescaling. Smooth-case points use numerical quadrature; Proposition 4.2 supplies the continuum approximation statement.

This is a factorization of an actual positive perturbed kernel, rather than an unconstrained error imposed on its factor. For the inverse factor $W_+ = K_+^{-1}$, a building block of whitening and Wiener–Hopf solution formulas, the ratio is $e^{iB_r(y)}$.

At $\varepsilon = 0.02$ and $y = 0.01$, three moments give a sharp phase budget of approximately 0.024906 radians; 25 moments give 0.013024 radians. At $y = 0.1$, the corresponding budgets are 0.010340 and 0.000952 radians. For this phase-only witness the relative complex response difference is $2|\sin(B_r/2)|$. Over its scalar multiples, the largest circular discrepancy is $2\sin(\min(B_r, \pi)/2)$, a distinction needed when unwrapped error exceeds π.

An interior complex evaluation is an exponentially regularized frequency response, not an ordinary real-frequency measurement. The example does not certify a closed-loop controller, a real-signal filter with even spectral symmetry, or the solution of a complete physical problem.

8 Use and limitations

The immediate applications are selecting a calibration order or an interior evaluation height under a justified log-error budget, and generating adversarial positive symbols with known factor behavior. An engineering integration must obtain the band support, envelope, and moment tolerances from its own acquisition or approximation procedure.

The model assumes exact information outside the band. Uncertain external tails require additional envelopes and integrals. The centre laws use a constant envelope and unweighted polynomial moments. Matrix factorization does not generally reduce to commuting scalar logarithms. Even-spectrum constraints remove the centre-phase problem. Smoothness without a derivative budget preserves the supremum; a fixed regularity bound may not.

The off-centre implementation is tested at low moment orders. Exact algebra and root isolation may become expensive at larger orders. The certifier checks proposed functions rather than proving a third-party optimizer correct. This makes optimizer accuracy dispensable for soundness, but retains the explicit dependency on the algebraic and interval libraries.

Further work could optimize over experimentally available calibration functionals, rather than increasing a fixed polynomial family. Adding a derivative budget would replace the free sign extremizers by a different constrained control problem. Neither extension is resolved here.

9 Conclusion

Positive scalar kernels with bounded finite-band logarithmic uncertainty admit exact marginal factor-error laws under polynomial calibration. Chebyshev sign functions attain both laws, separately, and smooth compactly supported perturbations approach the same suprema. The phase budget separates exponential improvement at fixed complex height from the boundary sensitivity retained by every fixed moment order. Best-L^1 duality and checked primal–dual brackets extend the calculation beyond the closed-form centre case, providing an auditable accuracy interface for scalar Wiener–Hopf factorization.

Data and software availability

The artifact is available at `https://github.com/shi1720/phasebudgets`. Commands `python run_experiments.py`, `python make_figures.py`, and `python -m unittest discover -s tests -v` regenerate records, figures, tables, and tests. The saved environment uses Python 3.12.14, NumPy 2.5.4, SciPy 1.18.1, mpmath 1.3.0, and python-flint 0.9.0. The manifest records actual versions. Figures are generated from the formulas and saved measurements.

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